



XMBA 201A Managerial Economics

Lecture 1: Intro to Decision Making

Professor Steve Tadelis

1

Why study Economic Analysis?

- Success comes from two main sources:
 - Dumb luck!
 - Careful decision making
- Core deficit in the world at large: ***Critical Thinking!!***
- XMBA201A is ***not*** about dumb luck. It's about how economic tools help ***critical thinking*** and ***structured decision making***.
- How does economics help structure critical thinking?
(or how does a microeconomist think?)

2

Thinking like an Economist

- Carefully compare **costs and benefits**.
- These matter **at the margin** of a decision and **not “on average”**
- **Strategic actors** drive the data generating process (beware of ML!!)
- You get what you pay for: **Incentives matter!**

Caveats: Economics doesn't...

1. ...tell you what goal to achieve. It tells you how to achieve it.
2. ...answer “should I help Alice at the expense of Bob”? It uncovers tradeoffs.
3. ...deal with “gut feelings”. Ignore your gut at your own peril!

3

Applications

- Decision making in an uncertain world
- Identify relevant costs for each decision
- Set prices to maximize profits
- Packaging and versioning to increase profits
- Understand how competitive markets work
- Learn to engage in strategic situations
- Design incentives for success

4

Methodology

- Identify a (decision) problem
- Strip the complicated real world down to a simple, tractable “toy model” (economic framework)
- What’s the risk of simplifying reality with a model?

Garbage in → Garbage out!

“All models are wrong, some models are useful!” (G. Box)

⇒ **Think hard about assumptions; perform sensitivity analysis**

- Use cases and news articles to apply the learned reasoning

5

Not Your Typical Undergrad Microeconomics Class!

- Ask not what’s good for “the economy” but what’s good for you, the decision maker!
- Do only a little bit of competitive market analysis
 - You ***don’t*** want to be in a very competitive market where you have no pricing power and no barriers to entry (unless you have a real competitive advantage...)

6

Expectations

- **Name tags:** PLEASE! And it's easier for me if you sit in the same area
- **Class Participation – key!**
 - Come prepared – I will do some cold calling
 - Use your MBA voice; Step up and step back!
- **Ask Questions!**
 - Students always! (Be curious... Remember: there's no such thing as a stupid question!)
 - Question the Status Quo & confidence w/o attitude! (Disagree and commit...)
 - Beyond yourself! (Take one for the team...)
- **Technology in class: Minimal!** (no laptops/phones; flat tablets OK but no browsing)
- **AI use:** Your call – the exam is pencil & paper, so AI can be your friend or your worst enemy...
- **Academic Honesty and Grades** (all hail the curve!!)

7

Logistics

- **GSI (TA):** Simoni Jain
 - Office hours on Wednesdays 5-6pm at this zoom link - <https://berkeley.zoom.us/j/6567944609>
- **Contact me:** stadelis@Berkeley.edu
 - Usually reply within 24 hours
 - Nag me if that does not happen!
- **Submission of problem sets:** via bcourses
- **Class recordings:** available after class
- **Printed slide deck:** If you really want one, email Nathaniel or Ravi today

8

Fenmark Case:

9

Fenmark Restaurants: The Facts

Actions: how much fish to order

Outcomes: costs and revenues (profits)

Objective: profit maximization

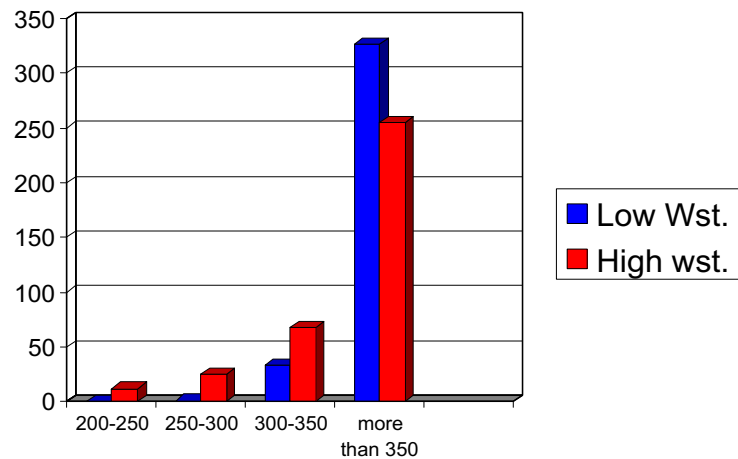
<u>Data</u>	<u>Low-Waste Restaurant</u>	<u>High-Waste Restaurant</u>
Price charged	\$15.00	\$15.00
Cost	\$6.00	\$6.00
Amount ordered	400	420
<u>Results</u>		
Average Daily Wastage	14.82	43.80
Average Daily Profit	\$3,377.67	\$3,123.00

- Gauss: “the data really point to our **throwing away too much fish.**”
- Question: Is there anything else in the data? What’s missing?

10

Demand (at $p = 15$)

- Someone has an easier job predicting demand...
- Is demand volatility playing an important role?



Professor Steve Tadelis

XMBA201A - Lecture 1

Page 11

11

What's the Objective?

- Fenmark should *maximize profits*
 - This includes all elements of profits (short run and long run; customers, reputation, etc.)
- Assuming that it's all about sales:
 - S : sales; D : demand; X : standing order
 - $Profits = (\$15) \times S - (\$6) \times X$ (note: $S = \min\{D, X\}$)

$$Profits = \begin{cases} (\$9) \times D - (\$6) \times (X - D) & \text{if } X \geq D \\ (\$9) \times D - (\$9) \times (D - X) & \text{if } X \leq D \end{cases}$$

Opportunity cost of waste (points to $(X - D)$ in the first case)

Opportunity cost of lost demand (points to $(D - X)$ in the second case)

Professor Steve Tadelis

XMBA201A - Lecture 1

Page 12

12

Counterfactual Analysis: Low Waste Restaurant

Data							
Price charged	\$	15.00					
Cost	\$	6.00					
Amount ordered		400					
				Hypothetical (-1)		Hypothetical (+1)	(sensitivity -5%)
							Dropped 8 days by 1
Results							
Average Daily Wastage		14.82					
Average Daily Profit	\$	3,377.67					
				$=\text{MIN}(B11,\$E\$4)*\$B\$2-\$E\$4*\$B\3		$=\text{IF}(B11=400,401,B11)$	
				\$ 3,375.96		\$ 3,379.38	\$ 3,379.04
Day	Sales	Wastage	Profit	Hypothetical (-1)		Hypothetical (+1)	
1	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
2	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
3	364	36	\$ 3,060.00	\$ 3,066.00	364	\$ 3,054.00	3054
4	348	52	\$ 2,820.00	\$ 2,826.00	348	\$ 2,814.00	2814
5	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
6	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
7	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
8	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
9	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
10	362	38	\$ 3,030.00	\$ 3,036.00	362	\$ 3,024.00	3024
11	400	0	\$ 3,600.00	\$ 3,591.00	401	\$ 3,609.00	3594
12	394	6	\$ 3,510.00	\$ 3,516.00	394	\$ 3,504.00	3504
13	330	70	\$ 2,550.00	\$ 2,556.00	330	\$ 2,544.00	2544

Professor Steve Tadelis

XMBA201A - Lecture 1

Page 13

13

Counterfactual Analysis: High Waste Restaurant

Data							
Price charged	\$	15.00					
Cost	\$	6.00					
Amount ordered		420					
				Hypothetical		Hypothetical (+1)	
Results							
Average Daily Wastage		43.80					
Average Daily Profit	\$	3,123.00					
				\$ 3,122.83		\$ 3,123.17	
Day	Sales	Wastage	Profit	Hypothetical		Hypothetical (+1)	
1	380	40	\$ 3,180.00	\$ 3,186.00	380	\$ 3,174.00	
2	394	26	\$ 3,390.00	\$ 3,396.00	394	\$ 3,384.00	
3	388	32	\$ 3,300.00	\$ 3,306.00	388	\$ 3,294.00	
4	288	132	\$ 1,800.00	\$ 1,806.00	288	\$ 1,794.00	
5	420	0	\$ 3,780.00	\$ 3,771.00	421	\$ 3,789.00	
6	376	44	\$ 3,120.00	\$ 3,126.00	376	\$ 3,114.00	
7	420	0	\$ 3,780.00	\$ 3,771.00	421	\$ 3,789.00	
8	178	242	\$ 150.00	\$ 156.00	178	\$ 144.00	
9	261	159	\$ 1,395.00	\$ 1,401.00	261	\$ 1,389.00	
10	411	9	\$ 3,645.00	\$ 3,651.00	411	\$ 3,639.00	

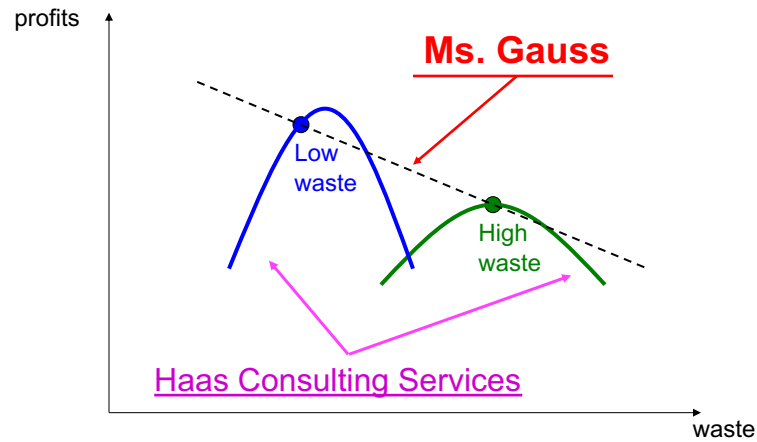
Professor Steve Tadelis

XMBA201A - Lecture 1

Page 14

14

What's going on?



15

Decision Analysis:

Formulation and Decision Trees

16

The Fundamental Trio: Description

- **A Decision Problem: Three Components:**
 1. **Actions:** What are my alternatives?
 2. **Outcomes:** What are the results of my choice?
 3. **Objectives:** What outcomes are desirable?
- **Examples of decision problems?**
 - What should I have for breakfast?
 - Which R&D project to pursue?
 - How to allocate marketing budget across channels?
 - Should I get an MBA or not?

17

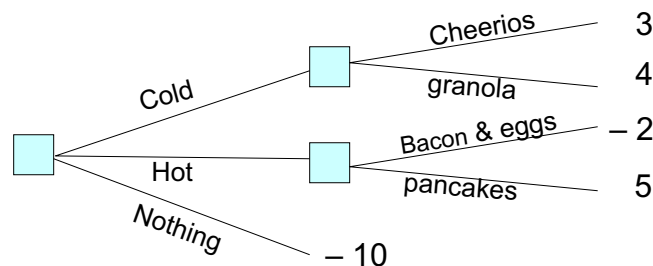
The Fundamental Trio: Details

- **Actions/Outcomes: problem specific**
 - Actions often follow previous actions over time
 - Outcomes are often uncertain
- **Objectives: What should these be?**
 - Firms: expected profit maximization
 - NFPs: bang for buck (same thing!)
 - Individual: expected “utility” maximization
 - This may seem cavalier, but it’s a very useful benchmark
 - “What I really need in order to feel fulfilled is _____”
 - Beware FOMO and adopt JOMO

18

A Useful Framework: Decision Trees

- **Static:** actions lead to (for now certain) payoffs
- **Dynamic:** actions lead to other possible actions, etc, until payoffs are realized.



Professor Steve Tadelis

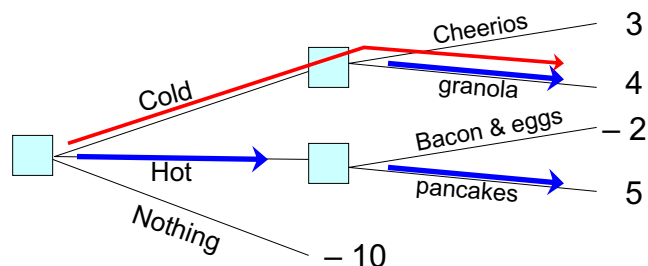
XMBA201A - Lecture 1

Page 19

19

An Algorithm: Backward Induction

- **Dynamics:** actions lead to other possible actions, so must look “forward” and solve “backward”:



- Decisions are ***irreversible!***

Professor Steve Tadelis

XMBA201A - Lecture 1

Page 20

20

Decision Analysis Under Uncertainty

Decision Trees with Probabilistic Events

Professor Steve Tadelis

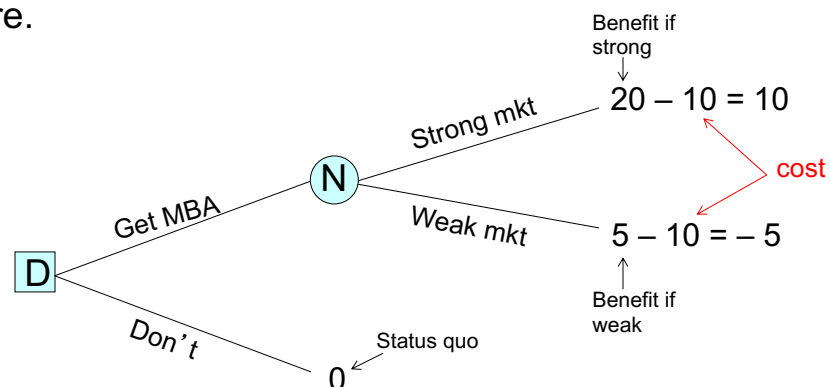
XMBA201A - Lecture 1

Page 21

21

An Example: Get an MBA?

- The value of an MBA depends on the strength of the market in the future.



- What should you do?

Professor Steve Tadelis

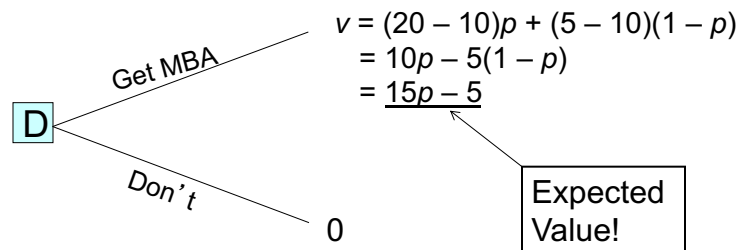
XMBA201A - Lecture 1

Page 22

22

Getting an MBA: A Simpler View

- If the probability of a strong market is p , then we can rewrite the decision tree:



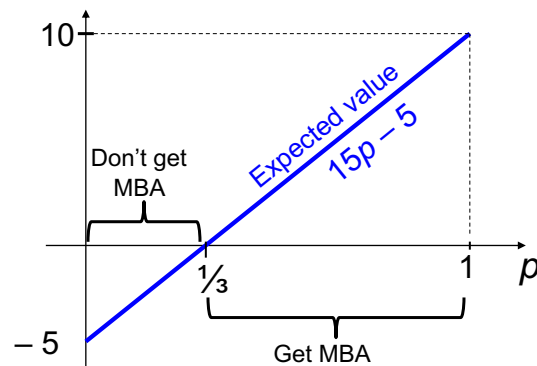
- So what should you do?
- Get an MBA if $15p - 5 > 0$, or $p > \frac{1}{3}$.

23

The Value of an MBA

- As p increases, the prospects are better, and an MBA is worth more:

- If $p > \frac{1}{3}$ then get MBA
- If $p < \frac{1}{3}$ then don't get MBA



**Don't confuse bad/good decisions
with bad/good outcomes!**

24

The Value of Information in Decision Analysis

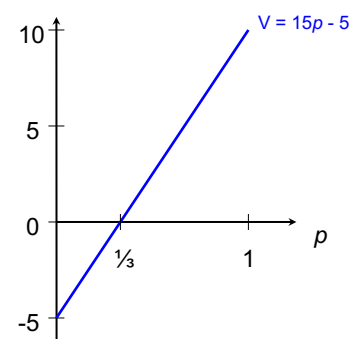
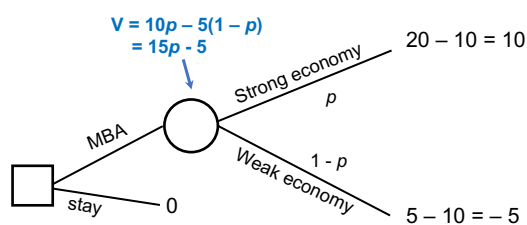
Professor Steve Tadelis

XMBA201A - Lecture 1

Page 25

25

Recall the Decision Tree for Getting an MBA



- What should you do?
- Depends on p :
 - Get MBA if and only if $15p - 5 > 0$
 - $\Leftrightarrow p > \frac{1}{3}!$

Professor Steve Tadelis

XMBA201A - Lecture 1

Page 26

26

A More Refined World

- Imagine that you **correctly** believe that $p = \frac{1}{2}$
 $\rightarrow \text{value} = 15 \times 0.5 - 5 = 2.5$ so you would get an MBA
- But the world is more refined:
 - If inflation is kept under control, chances of a strong economy are 90%
 - If inflation increases, the chance of a strong economy is 10%
 - Each “state” of inflation occurs with equal probability
- What does the “complete” decision tree look like?

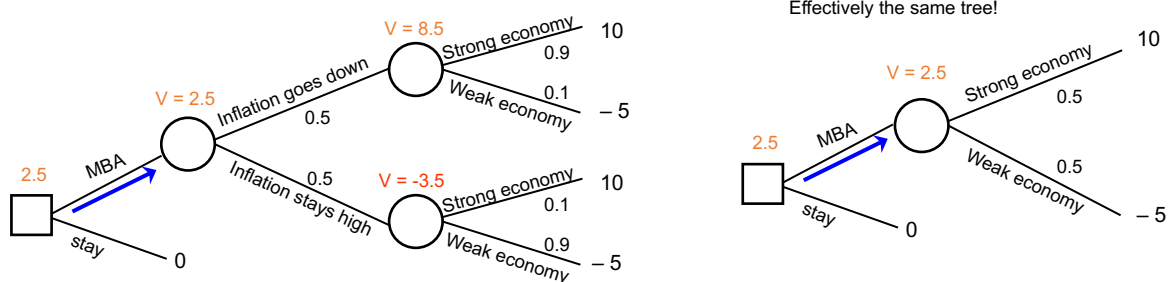
Professor Steve Tadelis

XMBA201A - Lecture 1

Page 27

27

A More Refined World



- You still need to decide on getting an MBA **without knowing** anything about whether inflation will increase or not. What do you do?
 - Because $p = \frac{1}{2} (> \frac{1}{3})$ you should get an MBA!
- Question:** If I could accurately predict whether inflation will increase or not, would you be interested in my information before making a decision ?
- How much would you be willing to pay for it?

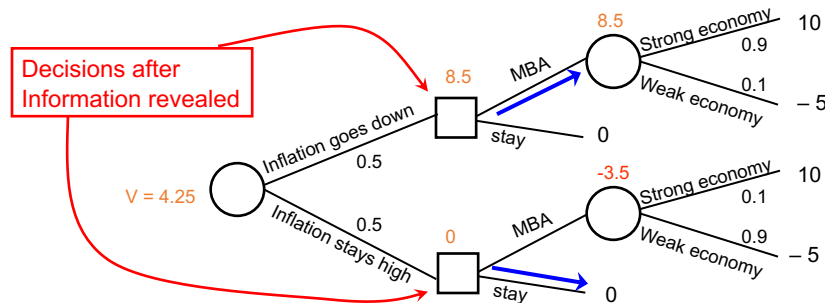
Professor Steve Tadelis

XMBA201A - Lecture 1

Page 28

28

The Value of Information: New Decision Tree



- Your decision can be made **contingent** on the information you learn!
- What is your **willingness to pay** for the information?
- The **added value** of the information:

$$\text{Price of info} \leq \text{value}_{\text{with}} - \text{value}_{\text{without}} = 4.25 - 2.5 = 1.75$$

The Value of Information: Rules

- The first rule of Information:
 - *Information that cannot affect your decision is worthless! Otherwise, it is valuable.*
- The second rule of Information:
 - *Information is most valuable when you are clueless!*
- The third rule of Information:
 - *Information is most valuable when without it, you are close to being indifferent between some alternatives.*

Summary: Decision Trees

- **Rigorous way** to capture actions, outcomes, uncertainty and payoffs
- Forces us to put structure on the problem **and identify assumptions!**
- Most challenging is choosing values for outcomes and probabilities for events (**assumptions!**)
- Structured way to **challenge assumptions** and consider **alternative scenarios** (sensitivity analysis)
- Irreversible actions means “solve” the decision tree backward
- Don’t confuse decision quality with outcomes quality! (luck...)
 - Pay attention to HOW decisions are made, not only to the decision itself
- When information changes the decision tree it is often valuable